

Ziqi CHEN

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Research Interests: Efficient & Adaptive LLM, Adaptive Computation, Agent Evolution

EDUCATION

The Chinese University of Hong Kong, Shenzhen (CUHKSZ)

Shenzhen, China

B.S. in Computer Science and Engineering

Sep. 2023 – May 2027 (Expected)

CGPA: 3.69/4.00 | GRE: 329 + 3.5 | TOEFL: 111

PUBLICATIONS & PREPRINTS

The Catfish Effect: Heterogeneous Competition in Group-Relative Learning

Yiyang Lu, **Ziqi Chen**, Xilin Xia, Hongyuan Zha.

Under review at ICLR. 2026.

DocMaster: A Hierarchical Structure-Aware System for Document Analysis

Ziqi Chen, Yingli Zhou, Fangyuan Zhang, Quanqing Xu, Chuanhui Yang, Yixiang Fang.

arXiv preprint. 2026. | [Paper Website](#)

RESEARCH EXPERIENCE

Research Intern, UChicago

Advisor: Prof. Kexin Pei and Prof. Tian Li

• **SciTestBench**

Apr. 2026 – Present

- Developed a comprehensive coding benchmark to evaluate LLM agents' ability of adversarial test generation on scientific software across 9 GitHub repositories covering 8 domains. Claude Opus 5.5 achieved a 24.3% pass rate.

Research Intern, CUHKSZ

Advisor: Prof. Hongyuan Zha

• **The Catfish Effect: Heterogeneous Competition in Group-Relative Learning** Mar. 2026 – Sep. 2026

- Developed Catfish GRPO, a group-relative RL framework that introduces a heterogeneous competitor through scalar outcomes only, reshaping advantage normalization and adaptive rollout allocation without demonstrations, distillation, or competitor trajectories in the student loss.
- Evaluated across Qwen/Llama models (3B–8B) and math/multi-hop QA tasks; improved accuracy +2.07 pp on GSM8K and 3.05 pp on MATH-500 over Vanilla GRPO, and showed that a virtual competitor can recover substantial gains without competitor inference.

Research Intern, CUHKSZ

Advisor: Prof. Benyou Wang and Dr. Yan H

• **Adaptive Routing for Multi-Timescale Memory in Language Models**

Feb. 2026 – Apr. 2026

- Proposed a token-level routing mechanism for dynamically selecting read and write depths in the Continuum Memory System (CMS) based on Nested Learning.
- Preliminary analysis of a four-layer CMS revealed non-uniform layer contributions across tokens, motivating adaptive computation for improved efficiency.

• **GRouter: A Learned Model Routing System for Multi-Agent Systems**

Dec. 2025 – Feb. 2026

- Developed a learned model routing system for multi-agent systems that assigns cost-effective base LLM

models based on the specific tasks while balancing cost and accuracy, by training a lightweight MLP router for real-time model selection with curated multi-LLM graph data

- Achieved a **4.2%** token cost reduction while preserving **90.3%** accuracy across GSM8K and HumanEval
- **MAS-Designer: Optimizing Multi-Agent Systems via GRPO** Oct. 2025 – Dec. 2025
 - Refined *G-Designer* by inducing GRPO and incorporating a utility-based reward function to stabilize the training process. Formulated multi-agent communication topology optimization as a graph pruning problem and solved it via GRPO.
 - Achieved a **9.7%** token cost reduction while preserving **89.1%** accuracy across GSM8K and HumanEval

Research Intern, CUHKSZ

Advisor: Prof. Yixiang Fang

- **DocMaster: A Hierarchical Structure-Aware Document Analysis System** May. 2025 – Sep. 2025
 - Designed an end-to-end system for document retrieval and filtering by combining constraint-based semantic clustering (PC-Kmeans + LLM-judged links) and hypergraph retrieval for cross-section semantics

INDUSTRIAL EXPERIENCE

Founding Engineer, Mortar

Dec. 2025 – Feb. 2026

Built a data-driven AI platform for real-world camera feed analysis at scale on Vercel and Supabase. Secured \$30,000 in startup funding from Anthropic and NVIDIA.

NLP Engineer, Lalamove

May 2025 – Aug. 2025

Fine-tuned a multilingual BERT model on QA dialogues, achieving **95.7% consistency** with existing services and saving approximately \$50,000 annually.

PROFESSIONAL SERVICE

Reviewer, ICLR 2026 Workshop on AI with Recursive Self-Improvement

Mar. 2026

Presenter, The 1st CUHK(SZ) Undergraduate Research Conference

Feb. 2025

AWARDS & HONORS

- UChicago Student Summer Research Fellowship (\$5000) Apr. 2026
- Undergraduate Research Award Mar. 2026
- Dean's List 2023 – 2026
- Undergraduate Student Teaching Fellows 2024 – 2026
- GBA International Programming Contest: First Prize Apr. 2025
- ShanHaiWoo 2025 Scholarship (\$1,000) Aug. 2025
- CUMCM (Contemporary Undergraduate Mathematical Contest in Modeling): Third Prize Sep. 2025
- ICM/MCM: Honorable Mention May 2025

SKILLS

- **Communities:** CAMEL Contributor, Sentient Ambassador, Obsidian Plugin Developer
- **Interests:** Reading, Travelling Table Tennis, Swimming